NLP 201: Finite State Automata

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Learning Objectives of NLP 201-203

- Ability to take a description of a model or system and implement it
- Ability to read current papers, understand them, and implement them
- Knowledge of lots of methods for solving problems (including older or alternative methods)
- Intuition into why the methods work and why they were developed
- Theoretical knowledge of why the methods work
- Debugging skills (difficult skill, *very important*)
- Ability to develop new solutions to problems

The plan for the next few lectures

For the next few lectures, we'll start with the simplest models of language.

- Finite state automata (FSAs, **today**)
- Regular languages
- Finite state transducers (FSTs)
- Later: Weighted FSAs and FSTs (WFSAs and WFSTs)

Why learn FSAs?

- Many models (HMMs, n-gram LMs, etc) are special cases of WFSAs
- FSAs and FSTs are used to model morphology
- First step in the formal language hierarchy
- Useful for understanding the modeling capacity of deep NN (for example, A Formal Hierarchy of RNN Architectures `https://arxiv.org/pdf/2004.08500.pdf`)
- Another tool for your toolbox
  - FSAs are incredibly fast

Goals of Today's Lecture

- Understand concepts of **formal languages**
- Understand **operations** on formal languages
- Refresher on proof techniques of **set containment** and **induction**
- Understand two FSAs: **deterministic finite automata** (DFAs), **nondeterministic finite automata** (NFAs) and their equivalence using the **powerset construction**.

Natural languages

- **Natural languages** are languages learned without explicit instruction by children from speakers in their environments
  - e.g., English, Mandarin, Hindi, Cantonese, German
- Natural languages link sounds or visual information (**linguistic forms**) with conceptual/intentional representations (**linguistic functions**)

Linguistic Concept: Grammar

Native speakers have a **grammar**, a (infinite) set of sentences that they accept as being "well-formed" **a sentence can either be in the speaker's grammar, or not in the speaker's grammar**

Example:

- They says "hello."
- They say "hello."

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What about:

- Colorless green ideas sleep furiously. **Another example: "Colorless green idea sleep furiously." doesn't work, but "A colorless green idea sleeps furiously" does.**

Formal languages

- **Formal languages** are (usually) infinite sets of strings described mathematically
- Formal languages may be used to describe things that are not natural languages, e.g. computer languages

Formal languages

- In this lecture, I will use the terms
  - **alphabet**, the symbols in the language
  - **word**, a string of symbols

Formal languages

- In this lecture, I will use the terms
  - **alphabet**, the symbols in the language
  - **word**, a string of symbols
- Could also have used the terms
  - **vocabulary**, the symbols in the language
  - **sentence**, a string of symbols

Fundamental definitions

Σ is a finite, nonempty **alphabet**

Each symbol is σ in the set is a "letter"

Usually denoted by lowercase letters $a, b, c, \ldots$

A **word** (sentence, string) $\boldsymbol{w} = w_1 w_2 \ldots w_n$

Usually denoted by bold, italic lowercase $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \ldots$

$|\boldsymbol{x}|$ is the **length** of word $\boldsymbol{x}$

ε is the **empty word**, and $|\varepsilon| = 0$

A **language** L is a *set* (finite or infinite) of words from a given alphabet Σ

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$|\boldsymbol{x}|$ is

ε is the **empty word**, and $|\varepsilon| = 0$

Careful!

$L_1 = \{\varepsilon\}$ is **not the same** as $L_2 = \emptyset$

A **language** L is a *set* (finite or infinite) of words from a given alphabet Σ

Formal language operations

Set theoretic **operations** apply to languages

$$L_1 \cup L_2 = \{\boldsymbol{w} \in \Sigma^* \mid \boldsymbol{w} \in L_1 \vee \boldsymbol{w} \in L_2\}$$ **union**

$$L_1 \cap L_2 = \{\boldsymbol{w} \in \Sigma^* \mid \boldsymbol{w} \in L_1 \wedge \boldsymbol{w} \in L_2\}$$ **intersection**

$$\overline{L}\ (= L^C) = \{\boldsymbol{w} \in \Sigma^* \mid \boldsymbol{w} \notin L\}$$ **complementation**

$$\left.\begin{aligned} \overline{(L_1 \cup L_2)} &= \overline{L_1} \cap \overline{L_2} \\ \overline{(L_1 \cap L_2)} &= \overline{L_1} \cup \overline{L_2} \end{aligned}\right\}$$ **de Morgan's laws**

Concatenation

Concatenation: denoted by a decimal point (but often omitted when meaning is clear from context)

String concatenation

$\boldsymbol{x} = aa \quad \boldsymbol{y} = bb \quad \boldsymbol{x}.\boldsymbol{y} = \boldsymbol{x}\boldsymbol{y} = aabb$

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$L_1 = \{a, b\} \quad L_2 = \{c, d\} \qquad L_1L_2 =$

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$L.\{\varepsilon\} = L$

Power operation

Power on letters/strings

$$\boldsymbol{w}^n = \boldsymbol{w}.\boldsymbol{w}^{n-1} \quad \boldsymbol{w}^0 = \varepsilon$$

$$\boldsymbol{w} = ab$$

$$\boldsymbol{w}^2 = abab$$

$$\ldots$$

Power on languages

$$L^n = L.L^{n-1} \quad L^0 = \{\varepsilon\}$$

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Power on languages

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The Kleene *

Kleene * on letters

$$a^* = \bigcup_{i=0}^{\infty} a^i = \{\varepsilon, a, aa, aaa, \ldots\}$$

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Kleene * on languages

$$L^* = \bigcup_{i=0}^{\infty} L^i = \{\varepsilon\} \cup L \cup L^2 \cup L^3 \ldots$$

$$= \{\boldsymbol{w} \mid \exists i \text{ s.t. } \boldsymbol{w} \in L^i\}$$

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Important notation

$\Sigma^* =$ set of all finite words over Σ

$\Sigma^i =$ set of all words of length i over Σ

The Kleene *

Kleene * on letters

$$a^* = \bigcup_{i=0}^{\infty} a^i = \{\varepsilon, a, aa, aaa, \ldots\}$$

Klee

Notational variant, the **Kleene** +

$$L^+ = \bigcup_{i=1}^{\infty} L^i = L \cup L^2 \cup L^3 \cup \ldots$$

Does not *add* the empty string ε

Important notation

$\Sigma^* =$ set of all finite words over Σ

$\Sigma^i =$ set of all words of length i over Σ

We stopped here.

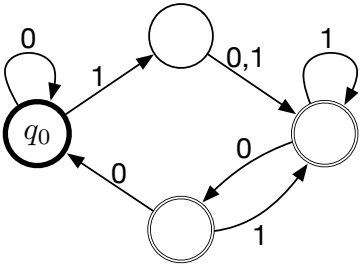
Definition: DFA

A **deterministic finite automaton** is a 5-tuple $M = \langle Q, \Sigma, \delta, q_0, F \rangle$ where

- Q is a finite **set of states**
- Σ is a finite **alphabet**
- $\delta : Q \times \Sigma \to Q$ is the **transition function**
- $q_0 \in Q$ is the **start (initial) state**
- $F \subseteq Q$ is the **set of final (accept) states**

$L(M) \subseteq \Sigma^*$ is **the language of** M, i.e. the set of strings M accepts

DFAs



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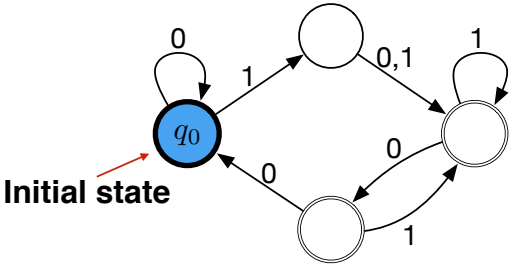

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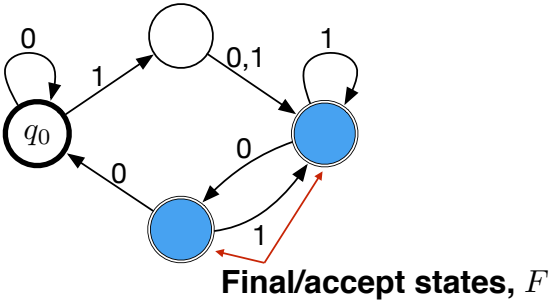

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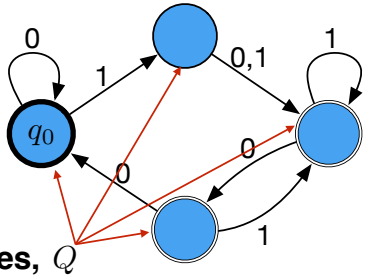
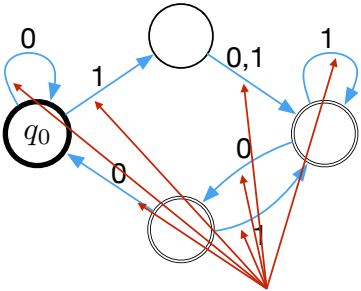


Set of states, Q

DFAs



Transition function ("edges"), δ

Acceptance

- M is a DFA with alphabet Σ
- An input sentence (word / string) ***w*** is a sequence $w_1 w_2 w_3 \ldots w_n$ where each $w_i \in \Sigma$
- M accepts ***w*** if after starting in q_0 and reading ***w*** it ends in an accept state, i.e., if there is a sequence of states $s_0, s_1, \ldots, s_n$ such that

$$s_0 = q_0$$
$$s_i = \delta(s_{i-1}, w_i)$$
$$s_n \in F$$

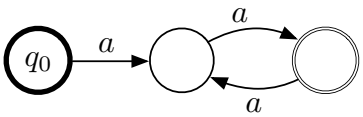
Acceptance and Analysis

- In many problems in natural language processing, we want to analyze a strings in terms of underlying operations.
- With finite state models, this usually corresponds to the question: what was the sequence of states $s_0, s_1, \ldots, s_n$ taken to produce some $\boldsymbol{w}$?

Example

The following DFA accepts the language

$L = \{a^{2n} \mid n \geq 1\}$



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~~$\delta : Q \times Q \to \Sigma$ is a **transition function**~~

$\delta : Q \times \Sigma \to 2^Q$ is the **transition relation**

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Definition: NFA

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Q is a finite **set of states**

An NFA accepts a word ***w*** if there exists a computation path that ends in a final state.

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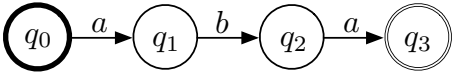
Remarks

- A DFA is required to have a **complete** transition function, whereas an NFA can have an **incomplete** transition function

- This is useful for specifying simple language generating automata, e.g. “linear chain” automata

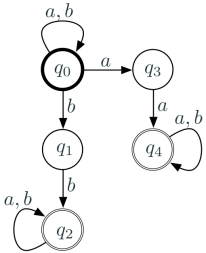


$$L(M) = \{aba\}$$

Board Work: NFA Example

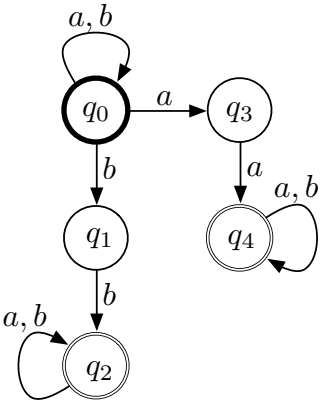
(5 Min) For the following NFA, does it accept the string $\mathbf{w} = abaa$? What sequence of states does it go through?

Can you describe in English the strings that it accepts?

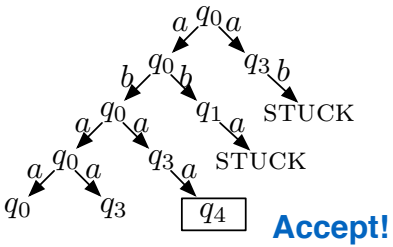


(3 Min) Discuss with a partner.

Example: NFA



Computation on $\boldsymbol{w} = abaa$



DFA & NFA Equivalence

Theorem. For every NFA A there exists a DFA A' s.t. $L(A) = L(A')$.

Regular Languages

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Example regular languages:

$$L = \Sigma^*$$

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Example non-regular languages (more later):

$$L = \{a^n b^n \mid n \geq 0\}$$

$$L = \{\boldsymbol{w} \mid \boldsymbol{w} \text{ is a grammatical sentence of English}\}$$

Generalization

It is helpful to generalize the transition function of DFAs in terms of **words** (instead of single symbols).

$$\delta(q, a) = q'$$

$$\hat{\delta} : Q \times \Sigma^* \to Q$$

$$\hat{\delta}(q, \varepsilon) = q$$

$$\hat{\delta}(q, \boldsymbol{x}\sigma) = \delta(\hat{\delta}(q, \boldsymbol{x}), \sigma)$$

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Formal definition of $L(M)$

$$L(M) = \left\{ \boldsymbol{w} \in \Sigma^* \mid \hat{\delta}(q_0, \boldsymbol{w}) \in F \right\}$$

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We also generalize in terms of **sets of states**:

$$P \in 2^Q$$

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Formal definition of $L(M)$

$$L(M) = \{\boldsymbol{w} \in \Sigma^* \mid \delta(q_0, \boldsymbol{w}) \cap F \neq \emptyset\}$$

We

$$\delta(P, \boldsymbol{x}) = \bigcup_{p \in P} \delta(p, \boldsymbol{x})$$

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DFA & NFA Equivalence

Theorem. For every NFA A there exists a DFA A' s.t. $L(A) = L(A')$.

Proofs

- Proofs in formal language theory often use **set containment** and **induction**

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- **Set containment**
  To prove that $(L_1 \cup L_2).L_3 = L_1.L_3 \cup L_2.L_3$

  one can show that (i)
  $\mathbf{w} \in (L_1 \cup L_2).L_3 \implies \mathbf{w} \in L_1.L_3 \cup L_2.L_3$

  and the other direction that (ii)
  $\mathbf{w} \in L_1.L_3 \cup L_2.L_3 \implies \mathbf{w} \in (L_1 \cup L_2).L_3$

Proofs

- Proofs in formal language theory often use **set containment** and **induction**

- **Induction**
  Induction is usually on the length of the word, i.e. to prove P is true for all $\mathbf{w} \in \mathbf{L}$, we:
  1. show P is true for words length 0 or 1
  2. hypothesize that P is true for all words $\leq$ length $n-1$
  3. prove that if P is true for all words $\leq$ length $n-1$, then P is true for for all words of length n

Board work: Proof by induction

On your whiteboards, take 5 min to do 1. and 3.
Then discuss with a partner for 3 min

- **Induction example**
  Prove that $\forall n \geq 0$

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$$

1. base: verify for $n = 0$
2. hypothesis: assume true for $n \leq k - 1$
3. inductive step: prove for $n = k$

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Proofs

- **Induction example**
  Prove that $\forall n \geq 0$

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$$

1. base: $n = 0$: $2^0 = 1 = 2^{0+1} - 1$
2. hypothesis: assume true for $n \leq k - 1$
3. inductive step: prove for $n = k$

$$\begin{aligned}
\sum_{i=0}^{k} 2^i &= 2^k + \sum_{i=0}^{k-1} 2^i \\
&= 2^k + 2^k - 1 \quad \text{(by inductive hypothesis)} \\
&= 2(2^k) - 1 \\
&= 2^{k+1} - 1 \quad \square
\end{aligned}$$

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Proof. *This is a constructive proof.* That is, given an NFA $A = \langle Q, \Sigma, \delta, q_0, F \rangle$ we construct a DFA $A' = \langle Q', \Sigma', \delta', q_0', F' \rangle$ s.t. $L(A) = L(A')$

DFA & NFA Equivalence

Theorem. For every NFA A there exists a DFA A' s.t. $L(A) = L(A')$.

Proof. *This is a constructive proof.* That is, given an NFA $A = \langle Q, \Sigma, \delta, q_0, F \rangle$ we construct a DFA $A' = \langle Q', \Sigma', \delta', q_0', F' \rangle$ s.t. $L(A) = L(A')$

Powerset construction.

$\Sigma' = \Sigma$

$Q' = 2^Q$ Constructed DFA states are **sets** of NFA states

$q_0' = \{q_0\}$

$F' = \{A \in Q' \mid A \cap F \neq \emptyset\}$

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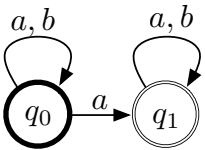
$\delta'(\emptyset, \sigma) = \emptyset \ \forall \sigma \in \Sigma'$ "Failure state"

$$\delta'(\{q_1, q_2, \ldots, q_i\}, \sigma) = \bigcup_{q \in \{q_1, q_2, \ldots, q_i\}} \delta(q, \sigma)$$

Example



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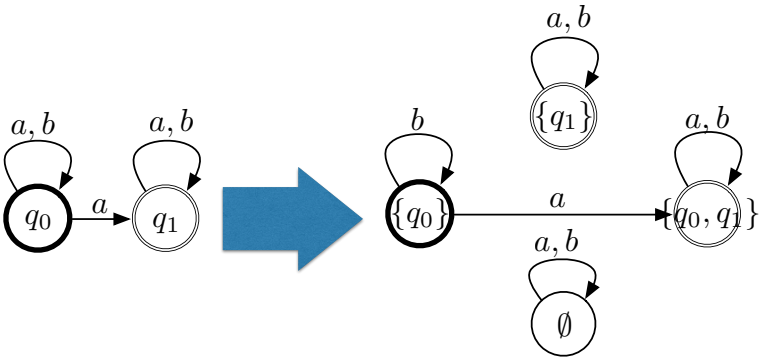

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Remarks

- NFAs therefore only accept regular languages
- NFAs and DFAs can be used interchangeably